

Library Book Management using Classes & Encapsulation

```
import java.util.Scanner;
```

```
class Book {  
    private int bookId;  
    private String title;  
    private String author;  
    private double price;  
    private boolean issued;  
  
    public Book(int bookId, String title, String author, double price) {  
        this.bookId = bookId;  
        this.title = title;  
        this.author = author;  
        this.price = price;  
        this.issued = false;  
    }  
  
    public int getBookId() {  
        return bookId;  
    }  
  
    public void issueBook() {  
        if (!issued) {  
            issued = true;  
            System.out.println("Book Issued Successfully.");  
        } else {  
            System.out.println("Book is Already Issued.");  
        }  
    }  
  
    public void returnBook() {  
        if (issued) {  
            issued = false;  
            System.out.println("Book Returned Successfully.");  
        } else {  
            System.out.println("Book is Already Available.");  
        }  
    }  
  
    public void display() {  
        System.out.println("-----");  
        System.out.println("Book ID : " + bookId);  
        System.out.println("Title : " + title);  
    }  
}
```

```
        System.out.println("Author : " + author);
        System.out.println("Price  : " + price);
        System.out.println("Status : " + (issued ? "Issued" : "Available"));
    }
}
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        Book[] books = new Book[100];
        int count = 0;
```

```
        while (true) {
```

```
            System.out.println("\n===== LIBRARY MENU =====");
            System.out.println("1. Add Book");
            System.out.println("2. Display Books");
            System.out.println("3. Search Book");
            System.out.println("4. Issue Book");
            System.out.println("5. Return Book");
            System.out.println("6. Exit");
```

```
            System.out.print("Enter Choice: ");
            int choice = sc.nextInt();
```

```
            switch (choice) {
```

```
                case 1:
```

```
                    System.out.print("Book ID: ");
                    int id = sc.nextInt();
                    sc.nextLine();
```

```
                    System.out.print("Title: ");
                    String title = sc.nextLine();
```

```
                    System.out.print("Author: ");
                    String author = sc.nextLine();
```

```
                    System.out.print("Price: ");
                    double price = sc.nextDouble();
```

```
books[count] = new Book(id, title, author, price);  
count++;
```

```
System.out.println("Book Added Successfully.");  
break;
```

case 2:

```
if (count == 0) {  
    System.out.println("No Books Available.");  
} else {  
    for (int i = 0; i < count; i++) {  
        books[i].display();  
    }  
}  
break;
```

case 3:

```
System.out.print("Enter Book ID: ");  
int searchId = sc.nextInt();
```

```
boolean found = false;
```

```
for (int i = 0; i < count; i++) {  
  
    if (books[i].getBookId() == searchId) {  
        books[i].display();  
        found = true;  
        break;  
    }  
}
```

```
if (!found)  
    System.out.println("Book Not Found.");
```

```
break;
```

case 4:

```
System.out.print("Enter Book ID: ");  
int issueId = sc.nextInt();
```

```
found = false;
```

```
for (int i = 0; i < count; i++) {  
  
    if (books[i].getBookId() == issuelId) {  
        books[i].issueBook();  
        found = true;  
        break;  
    }  
}  
  
if (!found)  
    System.out.println("Book Not Found.");  
  
break;
```

case 5:

```
System.out.print("Enter Book ID: ");  
int returnId = sc.nextInt();  
  
found = false;  
  
for (int i = 0; i < count; i++) {  
  
    if (books[i].getBookId() == returnId) {  
        books[i].returnBook();  
        found = true;  
        break;  
    }  
}  
  
if (!found)  
    System.out.println("Book Not Found.");  
  
break;
```

case 6:

```
System.out.println("Thank You");  
sc.close();  
return;
```

default:

```

        System.out.println("Invalid Choice.");
    }
}
}
}

```

OUTPUT:

```

===== LIBRARY MENU =====
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Choice: 1
Book ID: 101
Title: java
Author: no name
Price: 500
Book Added Successfully.

===== LIBRARY MENU =====
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Choice: 1
Book ID: 102
Title: python
Author: no name
Price: 400
Book Added Successfully.

===== LIBRARY MENU =====
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter Choice: 2
-----
Book ID : 101
Title   : java
Author  : no name
Price   : 500.0
Status  : Available
-----
Book ID : 102
Title   : python
Author  : no name
Price   : 400.0
Status  : Available

===== LIBRARY MENU =====
1. Add Book
2. Display Books
3. Search Book
4. Issue Book

```

2. Employee Payroll System using Inheritance

```
import java.util.Scanner;
```

```
class Employee {  
    int id;  
    String name;  
  
    Employee(int id, String name) {  
        this.id = id;  
        this.name = name;  
    }  
  
    double calculateSalary() {  
        return 0;  
    }  
  
    void display() {  
        System.out.println("\nEmployee ID : " + id);  
        System.out.println("Employee Name : " + name);  
        System.out.println("Salary      : " + calculateSalary());  
    }  
}
```

```
class PermanentEmployee extends Employee {  
  
    double basicSalary;  
    double hra;  
    double da;  
  
    PermanentEmployee(int id, String name, double basicSalary, double hra, double da) {  
        super(id, name);  
        this.basicSalary = basicSalary;  
        this.hra = hra;  
        this.da = da;  
    }  
  
    @Override  
    double calculateSalary() {  
        return basicSalary + hra + da;  
    }  
}
```

```
class ContractEmployee extends Employee {
```

```

int hoursWorked;
double ratePerHour;

ContractEmployee(int id, String name, int hoursWorked, double ratePerHour) {
    super(id, name);
    this.hoursWorked = hoursWorked;
    this.ratePerHour = ratePerHour;
}

@Override
double calculateSalary() {
    return hoursWorked * ratePerHour;
}
}

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Employee emp;

        System.out.println("==== Employee Payroll System =====");
        System.out.println("1. Permanent Employee");
        System.out.println("2. Contract Employee");

        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();

        System.out.print("Enter Employee ID: ");
        int id = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Employee Name: ");
        String name = sc.nextLine();

        if (choice == 1) {

            System.out.print("Enter Basic Salary: ");
            double basic = sc.nextDouble();

            System.out.print("Enter HRA: ");

```

```
double hra = sc.nextDouble();

System.out.print("Enter DA: ");
double da = sc.nextDouble();

emp = new PermanentEmployee(id, name, basic, hra, da);

} else if (choice == 2) {

    System.out.print("Enter Hours Worked: ");
    int hours = sc.nextInt();

    System.out.print("Enter Rate Per Hour: ");
    double rate = sc.nextDouble();

    emp = new ContractEmployee(id, name, hours, rate);

} else {

    System.out.println("Invalid Choice");
    sc.close();
    return;
}

System.out.println("\n----- Employee Details -----");
emp.display();

sc.close();
}
}
```

OUTPUT:


```
===== Employee Payroll System =====
```

```
1. Permanent Employee
```

```
2. Contract Employee
```

```
Enter Choice: 1
```

```
Enter Employee ID: 101
```

```
Enter Employee Name: john
```

```
Enter Basic Salary: 30000
```

```
Enter HRA: 25000
```

```
Enter DA: 200000
```

```
----- Employee Details -----
```

```
Employee ID   : 101
```

```
Employee Name : john
```

```
Salary       : 255000.0
```

3. Vehicle Rental System using Runtime Polymorphism

```
import java.util.Scanner;
```

```
class Vehicle {
```

```
    String vehicleNo;
```

```
    Vehicle(String vehicleNo) {  
        this.vehicleNo = vehicleNo;  
    }
```

```
    double calculateRent(int days) {  
        return 0;  
    }
```

```
    void displayBill(int days) {  
        System.out.println("\nVehicle Number : " + vehicleNo);  
        System.out.println("Days Rented   : " + days);  
        System.out.println("Total Rent   : Rs." + calculateRent(days));  
    }  
}
```

```
class Car extends Vehicle {
```

```
    Car(String vehicleNo) {  
        super(vehicleNo);  
    }
```

```
    @Override
```

```

    double calculateRent(int days) {
        return days * 1500;
    }
}

class Bike extends Vehicle {

    Bike(String vehicleNo) {
        super(vehicleNo);
    }

    @Override
    double calculateRent(int days) {
        return days * 500;
    }
}

class Bus extends Vehicle {

    Bus(String vehicleNo) {
        super(vehicleNo);
    }

    @Override
    double calculateRent(int days) {
        return days * 3000;
    }
}

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Vehicle vehicle;

        System.out.println("==== Vehicle Rental System =====");
        System.out.println("1. Car");
        System.out.println("2. Bike");
        System.out.println("3. Bus");

        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();
    }
}

```

```

        sc.nextLine();

        System.out.print("Enter Vehicle Number: ");
        String number = sc.nextLine();

        System.out.print("Enter Number of Rental Days: ");
        int days = sc.nextInt();

        if (choice == 1) {
            vehicle = new Car(number);
        } else if (choice == 2) {
            vehicle = new Bike(number);
        } else if (choice == 3) {
            vehicle = new Bus(number);
        } else {
            System.out.println("Invalid Choice");
            return;
        }

        System.out.println("\n----- Rental Bill -----");
        vehicle.displayBill(days);
    }
}

```

OUTPUT:

```

===== Vehicle Rental System =====
1. Car
2. Bike
3. Bus
Enter Choice: 1
Enter Vehicle Number: tn 12 ue 8374
Enter Number of Rental Days: 2

----- Rental Bill -----

Vehicle Number : tn 12 ue 8374
Days Rented    : 2
Total Rent     : Rs.3000.0

```

4. Bank Account System using Data Hiding

```
import java.util.Scanner;
```

```
class BankAccount {
```

```

private int accountNumber;
private String holderName;
private double balance;

public BankAccount(int accountNumber, String holderName, double balance) {
    this.accountNumber = accountNumber;
    this.holderName = holderName;
    this.balance = balance;
}

public void deposit(double amount) {
    if (amount > 0) {
        balance += amount;
        System.out.println("Amount Deposited Successfully.");
    } else {
        System.out.println("Invalid Deposit Amount.");
    }
}

public void withdraw(double amount) {
    if (amount <= 0) {
        System.out.println("Invalid Withdrawal Amount.");
    } else if (amount > balance) {
        System.out.println("Insufficient Balance.");
    } else {
        balance -= amount;
        System.out.println("Withdrawal Successful.");
    }
}

public void checkBalance() {
    System.out.println("\n----- Account Details -----");
    System.out.println("Account Number : " + accountNumber);
    System.out.println("Account Holder : " + holderName);
    System.out.println("Current Balance: Rs." + balance);
}

}

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

```

```
System.out.print("Enter Account Number: ");
int accNo = sc.nextInt();
sc.nextLine();
```

```
System.out.print("Enter Account Holder Name: ");
String name = sc.nextLine();
```

```
System.out.print("Enter Initial Balance: ");
double balance = sc.nextDouble();
```

```
BankAccount account = new BankAccount(accNo, name, balance);
```

```
while (true) {
```

```
    System.out.println("\n===== BANK MENU =====");
    System.out.println("1. Deposit");
    System.out.println("2. Withdraw");
    System.out.println("3. Balance Inquiry");
    System.out.println("4. Exit");
```

```
    System.out.print("Enter Choice: ");
    int choice = sc.nextInt();
```

```
    switch (choice) {
```

```
        case 1:
```

```
            System.out.print("Enter Deposit Amount: ");
            double dep = sc.nextDouble();
            account.deposit(dep);
            break;
```

```
        case 2:
```

```
            System.out.print("Enter Withdrawal Amount: ");
            double wd = sc.nextDouble();
            account.withdraw(wd);
            break;
```

```
        case 3:
```

```
            account.checkBalance();
            break;
```

```
        case 4:
```

```
            System.out.println("Thank You");
            sc.close();
```

```

        return;

    default:
        System.out.println("Invalid Choice");
    }
}
}
}
}

```

OUTPUT:

```

Enter Account Number: 75757
Enter Account Holder Name: john
Enter Initial Balance: 5000

```

```

===== BANK MENU =====

```

1. Deposit
2. Withdraw
3. Balance Inquiry
4. Exit

```

Enter Choice: 1

```

```

Enter Deposit Amount: 10000

```

```

Amount Deposited Successfully.

```

```

===== BANK MENU =====

```

1. Deposit
2. Withdraw
3. Balance Inquiry
4. Exit

```

Enter Choice: 3

```

```

----- Account Details -----

```

```

Account Number : 75757

```

```

Account Holder : john

```

```

Current Balance: Rs.15000.0

```

```

===== BANK MENU =====

```

1. Deposit
2. Withdraw
3. Balance Inquiry
4. Exit

5. University Admission System using Method Overloading

```

import java.util.Scanner;

```

```

class Admission {

```

```

    // Undergraduate Student

```

```

void calculateFee(double fee) {
    System.out.println("\nStudent Type : Undergraduate");
    System.out.println("Admission Fee: Rs." + fee);
}

// Postgraduate Student
void calculateFee(double fee, double labFee) {
    System.out.println("\nStudent Type : Postgraduate");
    System.out.println("Admission Fee: Rs." + (fee + labFee));
}

// Scholarship Student
void calculateFee(double fee, double scholarship, boolean isScholarship) {
    if (isScholarship) {
        System.out.println("\nStudent Type : Scholarship Student");
        System.out.println("Admission Fee: Rs." + (fee - scholarship));
    } else {
        System.out.println("Scholarship Not Applicable");
    }
}

}

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        Admission ad = new Admission();

        System.out.println("===== University Admission System =====");
        System.out.println("1. Undergraduate");
        System.out.println("2. Postgraduate");
        System.out.println("3. Scholarship Student");

        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();

        switch (choice) {

            case 1:

                System.out.print("Enter Undergraduate Fee: ");
                double ugFee = sc.nextDouble();

```

```

        ad.calculateFee(ugFee);
        break;

    case 2:

        System.out.print("Enter Course Fee: ");
        double pgFee = sc.nextDouble();

        System.out.print("Enter Lab Fee: ");
        double labFee = sc.nextDouble();

        ad.calculateFee(pgFee, labFee);
        break;

    case 3:

        System.out.print("Enter Course Fee: ");
        double fee = sc.nextDouble();

        System.out.print("Enter Scholarship Amount: ");
        double scholarship = sc.nextDouble();

        ad.calculateFee(fee, scholarship, true);
        break;

    default:

        System.out.println("Invalid Choice");
    }

    sc.close();
}
}

```

OUTPUT:

```

===== University Admission System =====
1. Undergraduate
2. Postgraduate
3. Scholarship Student
Enter Choice: 1
Enter Undergraduate Fee: 150000

Student Type : Undergraduate
Admission Fee: Rs.150000.0

```


6. Shape Area Calculator using Abstract Class & Polymorphism

```
import java.util.Scanner;
```

```
abstract class Shape {
```

```
    abstract double area();
```

```
    void display() {
```

```
        System.out.println("Area = " + area());
```

```
    }
```

```
}
```

```
class Circle extends Shape {
```

```
    double radius;
```

```
    Circle(double radius) {
```

```
        this.radius = radius;
```

```
    }
```

```
    @Override
```

```
    double area() {
```

```
        return 3.14 * radius * radius;
```

```
    }
```

```
}
```

```
class Rectangle extends Shape {
```

```
    double length, breadth;
```

```
    Rectangle(double length, double breadth) {
```

```
        this.length = length;
```

```
        this.breadth = breadth;
```

```
    }
```

```
    @Override
```

```
    double area() {
```

```
        return length * breadth;
```

```
    }
```

```
}
```

```
class Triangle extends Shape {
```

```
    double base, height;
```

```

Triangle(double base, double height) {
    this.base = base;
    this.height = height;
}

@Override
double area() {
    return 0.5 * base * height;
}
}

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Shape shape;

        System.out.println("==== Shape Area Calculator =====");
        System.out.println("1. Circle");
        System.out.println("2. Rectangle");
        System.out.println("3. Triangle");

        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();

        switch (choice) {

            case 1:

                System.out.print("Enter Radius: ");
                double r = sc.nextDouble();

                shape = new Circle(r);
                break;

            case 2:

                System.out.print("Enter Length: ");
                double l = sc.nextDouble();

                System.out.print("Enter Breadth: ");

```

```

        double b = sc.nextDouble();

        shape = new Rectangle(l, b);
        break;

    case 3:

        System.out.print("Enter Base: ");
        double base = sc.nextDouble();

        System.out.print("Enter Height: ");
        double height = sc.nextDouble();

        shape = new Triangle(base, height);
        break;

    default:

        System.out.println("Invalid Choice");
        sc.close();
        return;
    }

    System.out.println("\n----- Result -----");
    shape.display();

    sc.close();
}
}

```

OUTPUT:

```

===== Shape Area Calculator =====
1. Circle
2. Rectangle
3. Triangle
Enter Choice: 1
Enter Radius: 15

----- Result -----
Area = 706.5

```

7. Hospital Management System using Interfaces

```
import java.util.Scanner;
```

```
interface MedicalRecord {
```

```

void addRecord();

void displayRecord();
}

class Patient implements MedicalRecord {

    int patientId;
    String patientName;
    String disease;

    Scanner sc = new Scanner(System.in);

    public void addRecord() {

        System.out.print("Enter Patient ID: ");
        patientId = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Patient Name: ");
        patientName = sc.nextLine();

        System.out.print("Enter Disease: ");
        disease = sc.nextLine();
    }

    public void displayRecord() {

        System.out.println("\n----- Patient Record -----");
        System.out.println("Patient ID : " + patientId);
        System.out.println("Patient Name : " + patientName);
        System.out.println("Disease : " + disease);
    }
}

class Doctor implements MedicalRecord {

    int doctorId;
    String doctorName;
    String specialization;

    Scanner sc = new Scanner(System.in);

```

```

public void addRecord() {

    System.out.print("Enter Doctor ID: ");
    doctorId = sc.nextInt();
    sc.nextLine();

    System.out.print("Enter Doctor Name: ");
    doctorName = sc.nextLine();

    System.out.print("Enter Specialization: ");
    specialization = sc.nextLine();
}

public void displayRecord() {

    System.out.println("\n----- Doctor Record -----");
    System.out.println("Doctor ID    : " + doctorId);
    System.out.println("Doctor Name   : " + doctorName);
    System.out.println("Specialization : " + specialization);
}
}

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        MedicalRecord record;

        System.out.println("===== Hospital Management System =====");
        System.out.println("1. Patient");
        System.out.println("2. Doctor");

        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();

        if (choice == 1) {

            record = new Patient();

        } else if (choice == 2) {

            record = new Doctor();

```

```

    } else {

        System.out.println("Invalid Choice");
        return;
    }

    record.addRecord();
    record.displayRecord();
}
}

```

OUTPUT:

```

===== Hospital Management System =====
1. Patient
2. Doctor
Enter Choice: 1
Enter Patient ID: 1
Enter Patient Name: john
Enter Disease: heart attack

----- Patient Record -----
Patient ID   : 1
Patient Name : john
Disease      : heart attack

```

8. Online Shopping Order System using Packages

```
import java.util.Scanner;
```

```

class Product {
    int productId;
    String productName;
    double price;

    Product(int productId, String productName, double price) {
        this.productId = productId;
        this.productName = productName;
        this.price = price;
    }
}

```

```

class Order {
    Product product;
    int quantity;

    Order(Product product, int quantity) {

```

```

        this.product = product;
        this.quantity = quantity;
    }

    double calculateTotal() {
        return product.price * quantity;
    }

    void displayBill() {
        System.out.println("\n----- Order Bill -----");
        System.out.println("Product ID : " + product.productId);
        System.out.println("Product Name : " + product.productName);
        System.out.println("Price      : Rs." + product.price);
        System.out.println("Quantity   : " + quantity);
        System.out.println("Total Amount : Rs." + calculateTotal());
    }
}

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Product ID: ");
        int id = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Product Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Product Price: ");
        double price = sc.nextDouble();

        System.out.print("Enter Quantity: ");
        int qty = sc.nextInt();

        Product p = new Product(id, name, price);
        Order o = new Order(p, qty);

        o.displayBill();

        sc.close();
    }
}

```

```
}
```

OUTPUT:

```
Enter Product ID: 1
Enter Product Name: candy
Enter Product Price: 5
Enter Quantity: 10

----- Order Bill -----
Product ID   : 1
Product Name : candy
Price        : Rs.5.0
Quantity     : 10
Total Amount : Rs.50.0
```

9. Student Result Analyzer using Arrays of Objects

```
import java.util.Scanner;
```

```
class Student {
```

```
    int rollNo;
    String name;
    int mark1, mark2, mark3;
    int total;
    double average;
    char grade;
```

```
    Student(int rollNo, String name, int mark1, int mark2, int mark3) {
        this.rollNo = rollNo;
        this.name = name;
        this.mark1 = mark1;
        this.mark2 = mark2;
        this.mark3 = mark3;
    }
```

```
    void calculateResult() {
```

```
        total = mark1 + mark2 + mark3;
        average = total / 3.0;
```

```
        if (average >= 90)
            grade = 'A';
        else if (average >= 75)
            grade = 'B';
        else if (average >= 60)
            grade = 'C';
```



```

        else if (average >= 50)
            grade = 'D';
        else
            grade = 'F';
    }

    void display() {

        System.out.println("-----");
        System.out.println("Roll No : " + rollNo);
        System.out.println("Name   : " + name);
        System.out.println("Total   : " + total);
        System.out.println("Average : " + average);
        System.out.println("Grade   : " + grade);
    }
}

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Number of Students: ");
        int n = sc.nextInt();
        sc.nextLine();

        Student[] students = new Student[n];

        for (int i = 0; i < n; i++) {

            System.out.println("\nEnter Details of Student " + (i + 1));

            System.out.print("Roll No: ");
            int roll = sc.nextInt();
            sc.nextLine();

            System.out.print("Name: ");
            String name = sc.nextLine();

            System.out.print("Mark 1: ");
            int m1 = sc.nextInt();

            System.out.print("Mark 2: ");

```

```

        int m2 = sc.nextInt();

        System.out.print("Mark 3: ");
        int m3 = sc.nextInt();

        sc.nextLine();

        students[i] = new Student(roll, name, m1, m2, m3);
        students[i].calculateResult();
    }

    System.out.println("\n===== Student Results =====");

    int highest = students[0].total;
    int topperIndex = 0;

    for (int i = 0; i < n; i++) {

        students[i].display();

        if (students[i].total > highest) {
            highest = students[i].total;
            topperIndex = i;
        }
    }

    System.out.println("\n===== Class Topper =====");
    System.out.println("Name : " + students[topperIndex].name);
    System.out.println("Roll No : " + students[topperIndex].rollNo);
    System.out.println("Total : " + students[topperIndex].total);

    sc.close();
}
}

```

OUTPUT:

```
Enter Number of Students: 1
```

```
Enter Details of Student 1
```

```
Roll No: 1245432
```

```
Name: john
```

```
Mark 1: 90
```

```
Mark 2: 98
```

```
Mark 3: 9
```

```
===== Student Results =====
```

```
-----
```

```
Roll No : 1245432
```

```
Name    : john
```

```
Total   : 197
```

```
Average : 65.66666666666667
```

```
Grade    : C
```

```
===== Class Topper =====
```

```
Name : john
```

```
Roll No : 1245432
```

```
Total : 197
```

10. Smart Home Device Controller using Hierarchical Inheritance

```
import java.util.Scanner;
```

```
class Device {
```

```
    String deviceName;
```

```
    boolean status;
```

```
    Device(String deviceName) {  
        this.deviceName = deviceName;  
        status = false;  
    }
```

```
    void turnOn() {  
        status = true;  
        System.out.println(deviceName + " is ON");  
    }
```

```
    void turnOff() {  
        status = false;  
        System.out.println(deviceName + " is OFF");  
    }
```

```
    double powerConsumption() {
```

```

        return 0;
    }

    void display() {
        System.out.println("\nDevice Name : " + deviceName);
        System.out.println("Status    : " + (status ? "ON" : "OFF"));
        System.out.println("Power Consumption : " + powerConsumption() + " Watts");
    }
}

class Light extends Device {

    Light() {
        super("Light");
    }

    @Override
    double powerConsumption() {
        return 20;
    }
}

class Fan extends Device {

    Fan() {
        super("Fan");
    }

    @Override
    double powerConsumption() {
        return 75;
    }
}

class AirConditioner extends Device {

    AirConditioner() {
        super("Air Conditioner");
    }

    @Override
    double powerConsumption() {
        return 1500;
    }
}

```

```
}
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        Device device;
```

```
        System.out.println("==== Smart Home Device Controller =====");
```

```
        System.out.println("1. Light");
```

```
        System.out.println("2. Fan");
```

```
        System.out.println("3. Air Conditioner");
```

```
        System.out.print("Enter Choice: ");
```

```
        int choice = sc.nextInt();
```

```
        switch (choice) {
```

```
            case 1:
```

```
                device = new Light();
```

```
                break;
```

```
            case 2:
```

```
                device = new Fan();
```

```
                break;
```

```
            case 3:
```

```
                device = new AirConditioner();
```

```
                break;
```

```
            default:
```

```
                System.out.println("Invalid Choice");
```

```
                sc.close();
```

```
                return;
```

```
        }
```

```
        System.out.println("\n1. Turn ON");
```

```
        System.out.println("2. Turn OFF");
```

```
        System.out.print("Enter Operation: ");
```

```
        int op = sc.nextInt();
```

```
        if (op == 1)
            device.turnOn();
        else if (op == 2)
            device.turnOff();
        else
            System.out.println("Invalid Operation");

        device.display();

        sc.close();
    }
}
```

OUTPUT:

```
===== Smart Home Device Controller =====
1. Light
2. Fan
3. Air Conditioner
Enter Choice: 1

1. Turn ON
2. Turn OFF
Enter Operation: 1
Light is ON

Device Name : Light
Status      : ON
Power Consumption : 20.0 Watts
```